

Crank Powered Battery Replacement for Otoscopes

Alexandre Armengol-Urpi[†]

Department of Mechanical Engineering,
Massachusetts Institute of Technology,
Cambridge, MA 02139
armengol@mit.edu

Peter Duerst[†]

Department of Mechanical Engineering,
Massachusetts Institute of Technology,
Cambridge, MA 02139
pduerst@mit.edu

Jaemyon Lee[†]

Department of Electrical Engineering,
Massachusetts Institute of Technology,
Cambridge, MA 02139
jaemyon@mit.edu

Kerrie Wu[†]

Department of Mechanical Engineering
Massachusetts Institute of Technology,
Cambridge, MA 02139
kerriewu@mit.edu

Michael Fuenfer

Department of Pediatrics
Massachusetts General Hospital
Boston, MA 02114
mfuenfer@mgh.harvard.edu

Gim Hom

Department of Electrical Engineering,
Massachusetts Institute of Technology,
Cambridge, MA 02139
gim@mit.edu

Rachel Hoffman

Department of Mechanical Engineering,
Massachusetts Institute of Technology,
Cambridge, MA 02139
rachelmh@mit.edu

Nevan Hanumara

Department of Mechanical Engineering,
Massachusetts Institute of Technology,
Cambridge, MA 02139
hanamura@mit.edu

[†] †: Equally contributed, in alphabetical order by the last name of each author.

Alex Slocum

Department of Mechanical Engineering,
Massachusetts Institute of Technology,
Cambridge, MA 02139
slocum@mit.edu

ABSTRACT

Otoscopes are basic medical devices used for diagnosing diseases of the ear, such as common ear infections, which can lead to hearing loss. However, most otoscopes currently rely on grid electricity or disposable C-cell batteries, neither of which may be available in low-resource settings. To address this need, a crank-powered battery replacement for LED otoscopes was designed. The replacement occupies the same space as two series C-cell batteries and includes cathode and anode connections, making it operable without any modification to many commercially available otoscopes. Initial testing shows the battery replacement module can power > 50% brightness illumination (relative to operation with two C-cells) for at least 30 minutes with 2 minutes of comfortable crank-charging, and for at least 80 minutes with 5 minutes of crank-charging. This crank-rechargeable battery replacement could feasibly be used by clinicians in low-resource settings, and facilitate the improvement of healthcare in low-resource settings.

1. INTRODUCTION

Otoscopes are basic medical devices used to inspect the ear canal during routine medical examinations. They are critical to diagnosing and treating ear infections, which are common especially in young children and are a major cause of childhood hearing impairment. According to the World Health Organization, 360 million people worldwide have hearing impairment, 32 million of which are children, and 60 percent of childhood hearing loss is preventable [1].

Most conventional otoscopes rely on disposable cell batteries or grid electricity to charge rechargeable batteries. However, in low-resource settings, grid electricity is often unavailable or unreliable, and disposable batteries are either unaffordable or inaccessible. For example, 1.3 billion people worldwide [2], and 30% of health centers in Africa lack access to electricity [3,4]. Even in developed countries, natural disasters can cut off supply lines and electricity for extended periods of time. A hand-crank powered otoscope, which does not depend on disposable batteries or grid electricity, could help provide basic healthcare in these settings.

Various student projects have explored design of human-powered otoscopes and resulted in functional prototypes, but none have resulted in viable commercial products, and no current patents were found for crank-powered otoscopes [5-8]. However, many commercially available crank-powered flashlights and radios exist, which have greater operating power and duration than would be required for an otoscope. This suggests that a crank-powered otoscope is feasible.

Our approach to meet the need for an otoscope suitable for use in low-resource areas is the design of a hand-crank-powered otoscope battery replacement. The battery replacement module has the same shape and size of two series C-cell batteries, making it compatible with any LED otoscope originally powered by two C-cells. This module does not require modification of the otoscope and is a standalone product, requiring no additional parts. Operating an otoscope with the replacement module provides comparable brightness to operation with two C-cell batteries, and requires a few minutes of cranking to provide light for a full day's worth of patient examinations.

2. MATERIALS AND METHODS

The device presented in this work consists of a crank-rechargeable module to power most LED otoscopes that work with 2 C-cell batteries, shown in Figure 1. The crank can fold into the side of the device, providing it with the desired form factor to be inserted in the otoscope handle. In order to recharge the module, the crank can unfold to the cranking position. The module has an anode and a cathode as 2 C-cell batteries stacked in series would have.



Figure 1 The battery replacement module (left) next to two C-cell batteries (middle) and a commercial LED otoscope, which was designed to be operated with two C-cell batteries. The battery replacement module has the same size and electrical connections as the two series C-cell batteries, making it compatible with otoscopes like the one pictured without modification.

This section is organized as follows: section 2.1 will go through the functional requirements of the device, both at the electrical and at the mechanical level. Then, section 2.2 will explain how the previously presented electrical functional requirements are fulfilled, exposing the electrical architecture of the device. Finally, section 2.3 will present the mechanical architecture of the device, with its different components and materials, how it is operated and its durability.

2.1. Functional Requirements

2.1.1. Energy consumption and Energy generation

The generated energy must be higher than the consumed energy by the LED and the other parts; thus, there is a relationship:

$$Efficiency \times P_{generator} \times T_{generator} > P_{LED} \times T_{LED} \quad (1)$$

$$T_{generator} > \frac{P_{LED} \times T_{LED}}{Efficiency \times P_{generator}} \quad (2)$$

From the clinical survey, we found that clinicians use otoscopes for 1 minute per a patient, they meet 10 patients per hour and work for 8 hours; therefore, they use otoscopes for 80 minutes per day. To minimize cranking time with keeping the brightness of the LED, we should increase $P_{\text{generator}}$ and the efficiency of the system.

2.1.2. Lifetime of the rechargeable battery

Since the battery replacement module should last for many years of use, it is important to select a battery with high capacity and cycle lifetime. We chose Li-ion rechargeable battery because of its high energy density and comparable operation voltage with 2 C cells. Since each battery cycle is defined as the full-usage of the energy capacity, the lifetime is defined as

$$\text{Battery Lifetime} = \# \text{ of battery cycles} \times \text{days per battery cycle} \quad (3)$$

2.1.3. Safety of the battery

Although Li-ion battery has many advantages such as high energy density, it becomes explosive when it is overcharged. Also, once the Li-ion battery is overdischarged, it permanently loses rechargeability. Therefore, using a protection circuit is recommended for Li-ion batteries and we could choose Li-ion batteries having built-in protection circuit preventing overcharge and overdischarge.

2.1.4. Mechanical requirements

Since the module should last for many years of use, including during travel or in environments outside of typical clinician offices, it must be as durable as possible and able to withstand rough use and dropping. To eliminate the risk of misplacing any one part and therefore losing functionality of the entire module, the module is a single-piece standalone product with no detachable parts. Since the insert is fully contained inside the otoscope during use, the risk of dirt or water entering the module is low.

For maximum impact and ease of use, the module is compatible with most otoscopes using 2 C-cell batteries to power an LED light, without requiring modification to the otoscope. It must be comfortable and easy to typical human to charge the module.

2.2. Electrical architecture

The electrical architecture of the device can be divided into its two main modes: energy consumption and energy generation mode. These will be explained in the following sections.

2.2.1. Energy consumption

To make a battery replacement for the otoscope, key characteristics of the LED-based otoscope were measured. In the otoscope, the LED driving circuit was a series connection of a rheostat (variable resistor), an LED and 2 C cells. From the measurement, the forward voltage of the LED was 2.6 V (Figure 2), and the voltage of two C cells was about 3.05 V. At the maximum brightness setting, the built-in resistance of the otoscope including contact resistance and the rheostat was 14 ohm and the operation current was 30 mA. From this amount of current and daily usage from the survey (80 min/day), the charge consumption through the LED is 40 mAh / day, and this is the amount of charge we need to supply to the LED with a battery. In addition, most Li-ion batteries can be charged for 500 full-energy cycles.[9] To reach enough lifetime, 700 mAh, 3.7 V Li-ion battery from Zeasun was selected and the estimated lifetime with a standard daily usage was

$$\begin{aligned} \text{Battery Lifetime} &= \# \text{ of battery cycles} \times \text{days per battery cycle} \\ &= 500 \times (700 [\text{mAh}] / 40 [\text{mAh/day}]) = 8750 \text{ days} \end{aligned} \quad (4)$$

The estimated battery lifetime is longer than 20 years and we concluded that is more than enough for our purpose.

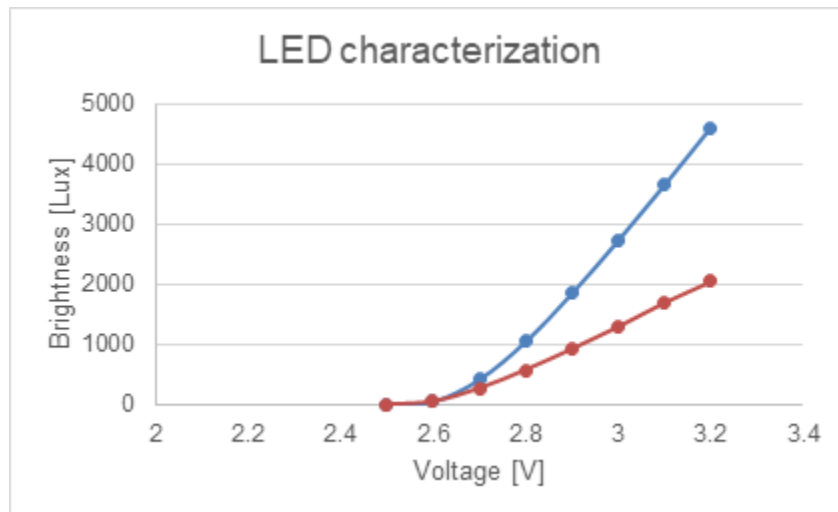


Figure 2 Characterization of LED in a series connection of a resistor, a battery and an LED. As the resistance changes (Blue: low R, red: high R), the brightness decreases, but the x-intercept, the forward voltage of LED, does not change.

Since battery, resistor (rheostat), and LED are connected in series and have the same current, the energy consumption (resistor and LED) and energy supply (battery) are proportional to the voltage (Figure 2). Since LED is a non-linear device having the same voltage during operation, the energy efficiency (V_{LED} / V_{Batt}) does not change regardless of the amount of current; thus, we added a series resistor to set the current same as the reference current using 2 C cells to get the same light intensity.

With 2 C cells, since $V_{LED} = 2.6$ V, the energy efficiency was 0.867 ($= 2.6/3$). We chose an Li-ion battery since it has many advantages as we mentioned before, but its operating voltage is 3.7-4.2 V and the energy efficiency became 0.62-0.70. The lower energy efficiency came from the mismatch between V_{LED} and V_{Batt} . There were two other methods to control LED brightness: pulse width modulation (PWM) and DC-DC conversion. PWM does not change energy efficiency due to the non-linearity of the LED (Figure 3-1) compared to using an additional resistor (Figure 3-2), but DC-DC converter can improve the energy efficiency (Figure 3-3). If the output current is about 10-30 mA, the energy efficiency of DC-DC conversion (from 3.7-4.2 V to 3 V) is about 90 % and the total energy efficiency could become $0.9 \times 0.867 = 0.78$; therefore, it is possible to improve the energy efficiency by 10-25 % with a DC-DC converter. Due to the size of DC-DC converter and capacitors, we did not use DC-DC converter in our first prototype. Once we use DC-DC converter instead of an additional resistor, we can get higher energy efficiency and also make the device compatible with most medical devices using 2 C cells.

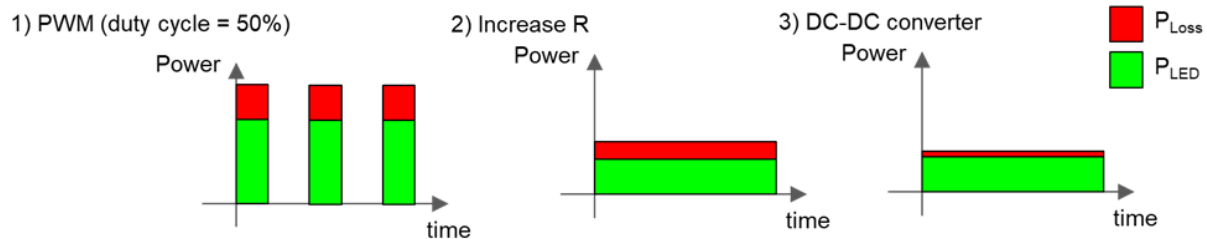


Figure 3. Energy consumption of the LED and energy loss diagram of various methods decreasing light intensity.

2.2.2. Energy generation

There are three key components in energy generation: the gear-motor generating voltage, a rectifier to avoid reverse current from the battery to the gearmotor, and the battery (Figure 4). Some commercialized crank-based flashlights and USB chargers used a single diode as a rectifier, but this single diode also blocks the current generated by left-handed people since they rotate the motor in reverse direction (counter clockwise) and generate negative voltage. To solve this problem, we chose a diode bridge (CBRHDSH2-40 from Central Semiconductor Corp.) as a rectifier, and both right-handed and left-handed people could generate and deliver current to the battery properly. Furthermore, to decrease the voltage loss of the rectifier, we chose Schottky diodes having lower forward voltage (0.4 V) rather than PN diodes having higher forward voltage (1.1 V). Although Schottky diode has lower breakdown voltage (40 V), our battery has lower output voltage (maximum voltage: 4.2 V) than the breakdown voltage; therefore, the breakdown of the diode bridge does not happen in our system and battery does not rotate the handle.

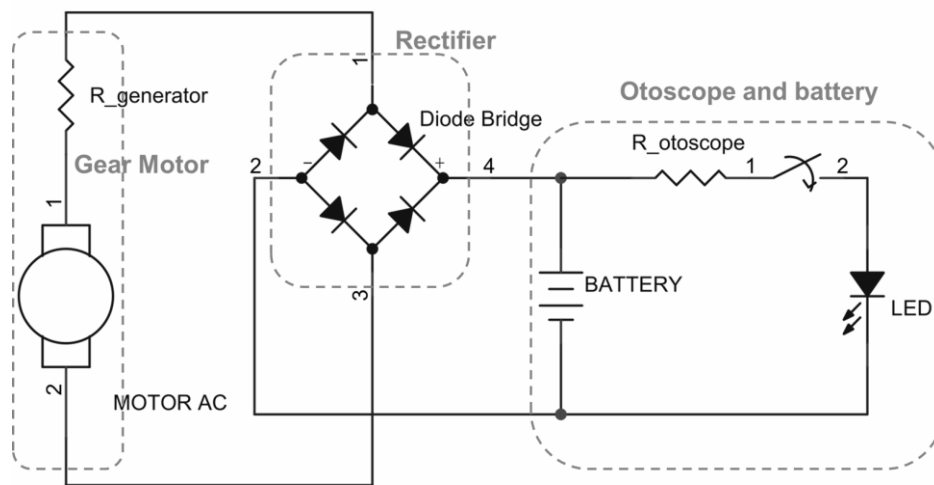


Figure 4. Schematic of our device and otoscope

The generated voltage was proportional to the angular velocity by Faraday's law. The generated current depends on 1) the voltage difference between the generated voltage by motor and the sum of the forward voltage of diode rectifier and the battery voltage, and 2) the system resistance including the resistance of coils in the motor. With an additional 1 ohm resistor for current measurement and the first prototype having a handle without an ergonomic knob, we found that the generated current was about

300 mA in average (Figure 5). The ratio between charging current and consuming current is 30:3, and the cranking time for 80 minute usage was about 8 minutes.

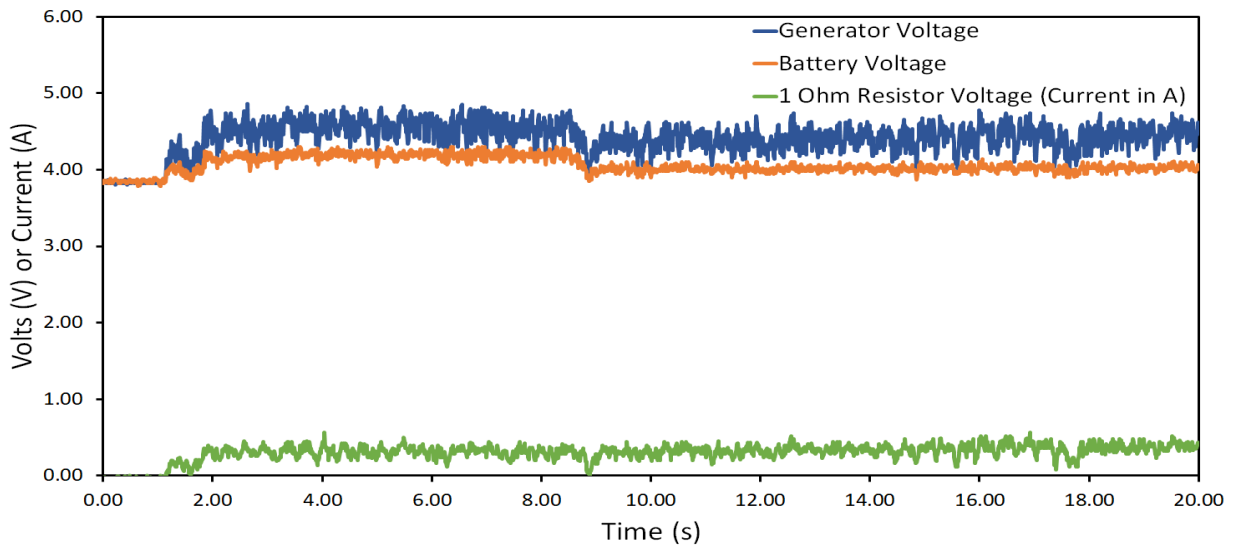


Figure 5. Bench-top test of charging operation. The average current was 300 mA (gear ratio = 78:1 and the diameter of the gearmotor = 20 mm)

2.3. Mechanical architecture

2.3.1. Components

A schematic of the device is shown in Figure 6. The mechanical architecture of the device consists of five main modules:

- a) *A motor with its gearbox.* A 6V brushed DC gear motor from Pololu [10] was used as the power generator. A gear ratio of 78:1 was selected to provide the most comfortable cranking experience while producing the desired voltage output.
- b) *A mounting hub (anode).* The hub interfaces with the motor shaft and the crank, and also acts as the anode of the device. An embedded hinge allows the crank to fold and unfold to switch between powering and cranking modes, respectively. Although the hinge can withstand most human forces by itself, the hub also includes a sturdy post that contacts the crank on the opposite side from the hinge (Figure 7). This design allows most of the cranking force to go through the post instead of the hinge during normal operation.

- c) *A crank*. The crank interfaces with the mounting hub through the embedded hinge and through the hub post. A crank radius of 46mm was selected based on space constraints and testing, which found this radius to be the most comfortable for cranking. It includes a rotating knob for more ergonomic and comfortable cranking, which also folds down to fit inside the body and maintain the cylindrical form factor.
- d) *A top cover (cathode)*. The cover is electrically connected to the cathode of the Li-ion rechargeable battery through resistor or DC-DC converter.
- e) *A 3D printed body*. The body encapsulates the components and gives the desired form factor to our device, allowing the crank to fold in when in powering mode.

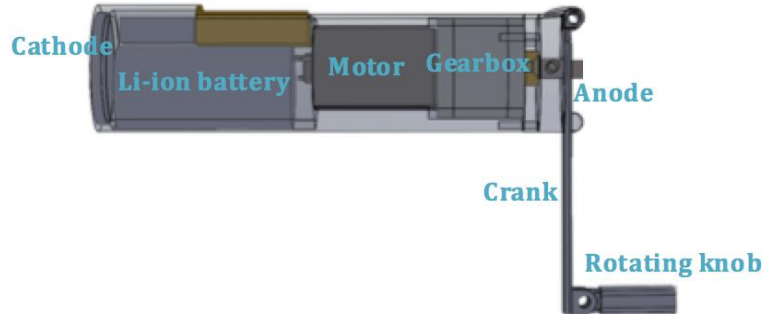


Figure 6. Schematic of the device in cranking position with its different components.

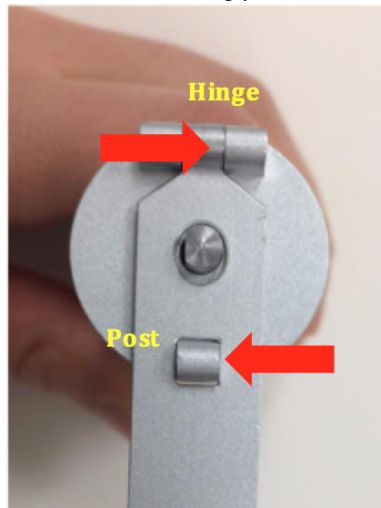


Figure 7. Anode side of the device with crank in recharging position. It can be seen how the the crank contacts the post to prevent forces from being transmitted through the hinge.

2.3.2. Operation

The device can be at two different states or modes: power mode with the crank folded, and charging mode with the crank unfolded (Figure 8). In the power mode, the form factor of the device is the same as 2 C-cell batteries so it can be inserted in the otoscope (or any other device working with 2 C-cell batteries stacked in series). In the charging mode, the crank is unfolded from its original position and placed at 90 degrees from the device axis, where it can be cranked. In this position, the post in the anode interfaces with the hole in the crank, as can be seen in Figure 9.



Figure 8. Crank stowed for energy consumption, and crank unfolded for energy generation.

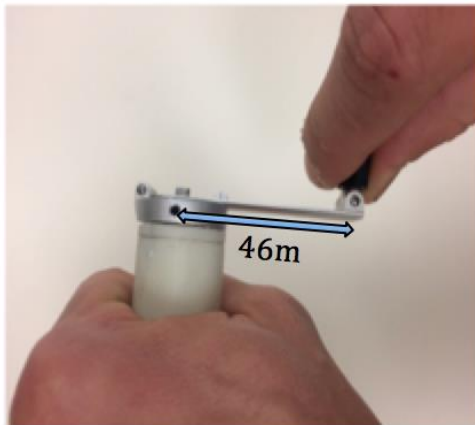


Figure 9. Cranking operation, with the crank arm at 90 degrees to the device axis and sitting on the anode of the device.

2.3.3. Manufacturing

The mounting hub, crank, and top cover were machined out of aluminum 6061, but future models may benefit from having parts machined out of steel since it would provide greater durability and strength, especially for critical parts such as the hinge. The dowel pin for the hinge was made out of alloy steel and press-fitted. The body and rotating knob were 3D printed.

2.3.4. Durability

There are several sensitive points where the cranking system could fail due to excessive forces or inappropriate device operation. Maximum forces and torques in most critical directions have been computed and most of them are high enough to confirm that the system will not break under normal operation and assuming the user does not intend to intentionally damage the device. However, the gearbox is a critical module since the gearmotor used in this prototype is not designed to be back-driven. Therefore, the maximum angular speed and acceleration before the gears break were computed.

The torque on the gearmotor shaft were modeled as having a component due to the inertia of the motor, which depends on the angular acceleration of the shaft, and a viscous component due to the current through the motor. The viscous component of the torque was calculated from a DC motor model:

$$\tau_v = k_t \cdot I \quad (5)$$

I is the charging current, which is traveling through the motor, and the current-to-torque constant $k_t = 0.172 \text{ N} \cdot \text{m}/\text{A}$ was calculated using the gearmotor specifications of stall torque and current [X].

The inertial component of the torque, τ_i , on the gearmotor shaft was calculated from an estimation of the polar moment of inertia of the motor's rotor:

$$\tau_i = GR^2 \cdot J \cdot \alpha \quad (6)$$

The gear ratio, $GR = 78.125$, is taken from the gearmotor's spec sheet, and α is the angular acceleration of the output shaft of the gearmotor. The moment of inertia, $J = 3.33 \times 10^{-7} \text{ kg} \cdot \text{m}^2$, of the

motor's rotor was computed by modeling the rotor as a cylinder rotating about its polar axis, with $\frac{1}{3}$ the mass (m) of the entire gearmotor and $\frac{2}{3}$ the radius (r) of the gearmotor:

$$J = \frac{1}{2} \cdot m \cdot r^2 \quad (7)$$

Plots obtained using the model are shown in Figure 10.

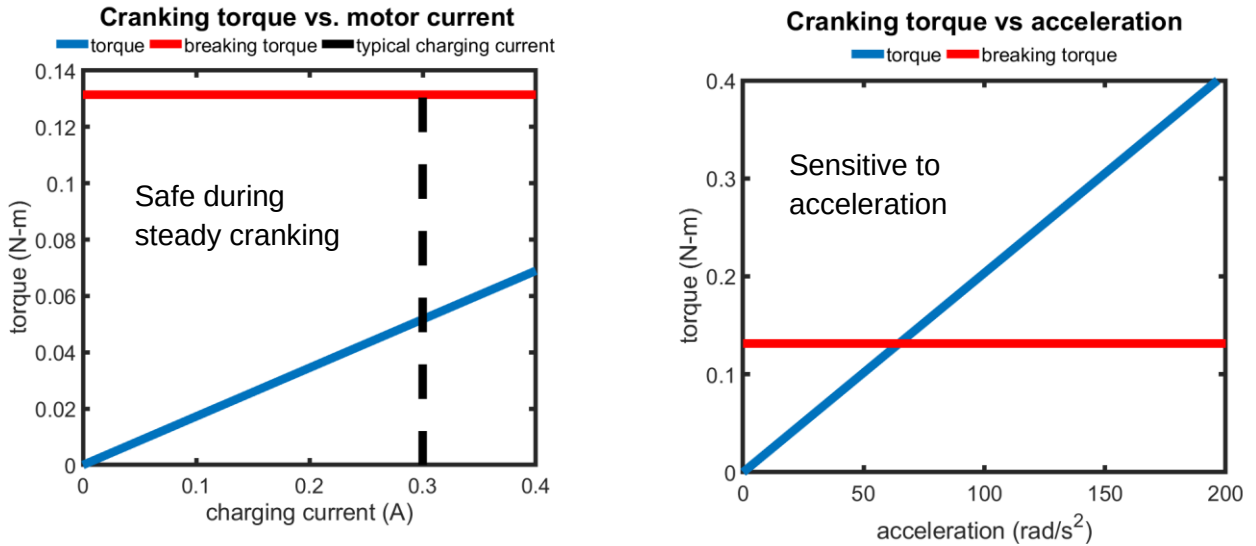


Figure 10 Plots representing the cranking and breaking torque obtained from calculations. Although the gears are strong enough to withstand steady cranking, high angular accelerations could break the gears.

It can be seen that under gentle and steady use the gearbox will not break, but it may be feasible damaging it with a strong user or erratic cranking that gives high angular acceleration. Improving this performance could be done in future models by designing a new gearbox, which would be more adequate for back-driving the motor.

3. RESULTS

3.1. Experimental Apparatus

The power module was tested using a otoscope set designed for use with two C-cell batteries and an LED light source. Worst-case scenario tests were performed, during which the module was cranked at a comfortable speed for 1 minute, 2 minutes, and 5 minutes from a fully discharged state. The light intensity at 1 cm away from the tip of the otoscope was measured in lux using a Vernier Light Sensor

[11], and compared to the intensity of the otoscope powered by 2 C-cell batteries. Voltage of the power module was measured using a digital multimeter.

3.2. Brightness and Charging

Operating the otoscope with 2-C-cell batteries produced a steady light intensity of about 5500 lux. The fully discharged battery replacement module had terminal voltages ranging from 2.62-2.71 V and produced no detectable light when inserted into the otoscope. After 1 minute of cranking, the replacement module voltage was 3.06 V, and the light intensity using the replacement module was only 1000 lux. However, after 2 minutes of cranking from discharge, the replacement module voltage was 3.20 V and the light intensity was about 3800 lux. Over the course of 30 minutes of continuous operation with no additional cranking, the light intensity decreased to about 3100 lux. A plot of the light intensity over time after 2 minutes of cranking is shown in Figure 11.

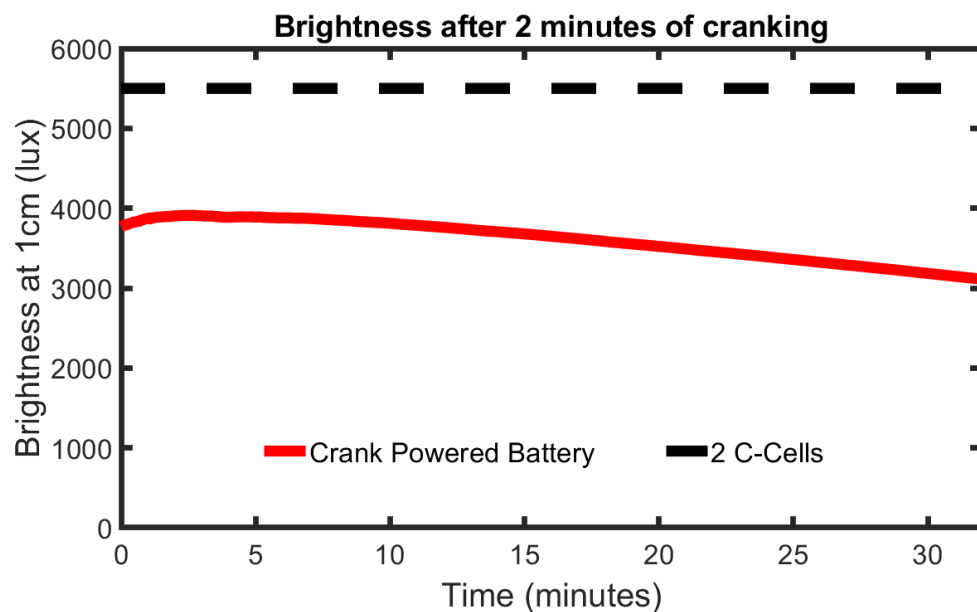


Figure 11 Comparison of the light intensity at 1 cm from the tip of the otoscope using the battery replacement module and using 2 C-cell batteries. Greater than 50% brightness is sustained for at least 30 minutes with 2 minutes of cranking from a discharged state.

After about 5 minutes of cranking from discharge, the replacement module voltage was 3.2 V, and the brightness was about 3900 lux, which decreased to about 2800 lux over 80 minutes. The light intensity over time is shown in Figure 12.

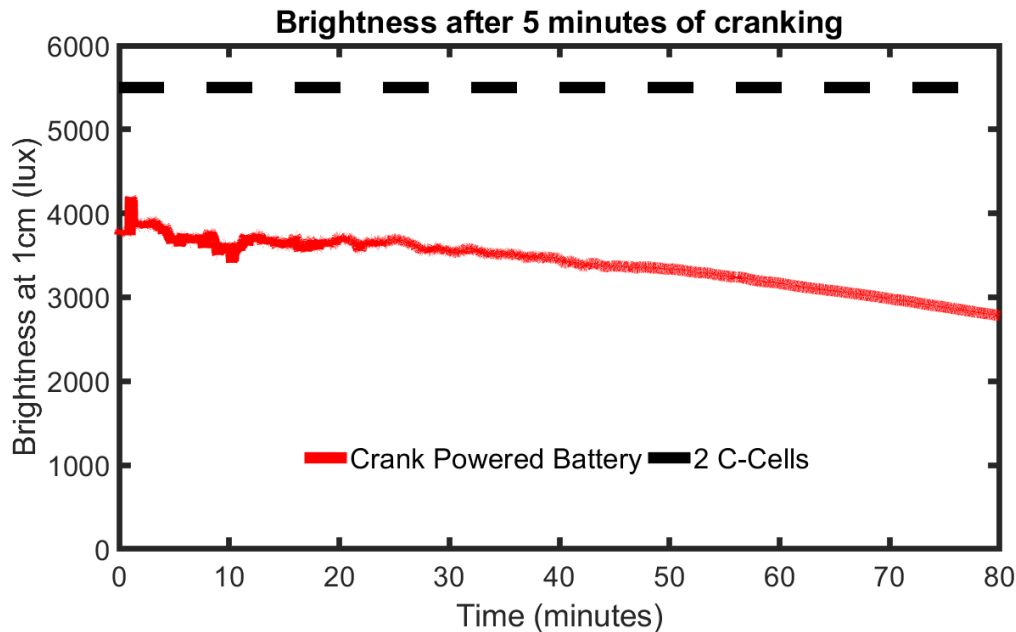


Figure 12. Using the battery replacement module, greater than 50% brightness (compared to operation with 2 C-cells) is sustained for at least 80 minutes with 5 minutes of cranking from a discharged state.

4. DISCUSSION

4.1. Brightness and Charging

After 2 minutes of cranking from a fully discharged state, it was possible to reach about 67% brightness relative to the brightness of the otoscope when powered by two C-cell batteries. The power module sustained > 50% brightness for greater than 30 minutes following.

From the estimation regarding daily otoscope usage in section 2.1.1, a total of 80 minutes of light would be required per day. Our worst-case testing indicates that our power module would allow for at least 80 minutes of 50% brightness illumination with 5 minutes of crank-charging, which is very feasible.

100% brightness illumination was not achieved in our worst-case tests because the battery replacement module voltage only reached 3.2 V, while the electronics were designed to provide 100%

brightness at 3.7 V. This illustrates the need for adding brightness regulation to the electrical circuit, and incorporating a better way of stepping down the battery voltage than using a series resistor.

Although our worst-case testing showed that we were not able to reach the same measured brightness, a physician noted that he did not notice a difference in brightness between the operating the otoscope with the power module and with two C-cell batteries. This may be because the human eye has a logarithmic response to brightness, rather than a linear response [12]. Future testing should be done to determine whether this brightness level is adequate, or if it should be increased.

4.2. Mechanical Design

Initial testing confirmed that the cranking force and speed needed to significantly charge the battery replacement module was comfortable for users. Users noted that the rotating knob on the crank was especially helpful for enhancing comfort during crank-charging.

Because the module requires no additional parts or modifications to the otoscope handle, it could be suitable for use in any device that uses two C-cells to power an LED. For example, the otoscope set used for testing the module also came with a compatible ophthalmoscope head, so our device could power both medical instruments without any further modification. Further testing regarding acceptable brightness levels would be required in order to validate using the insert with ophthalmoscopes or other devices, but it would not require any mechanical changes to the design.

4.3. Future work

As an initial next step, usability testing needs to be performed to determine what aspects need improvement, such as durability, comfort of use, and brightness. It would be ideal to have clinicians use the replacement module to power an otoscope and give feedback on what brightness level is acceptable during ear examinations.

The resistor used for stepping down the battery voltage should be replaced with a DC-DC converter in order to increase the battery replacement module's efficiency. A Power Management Integrated Circuit, or PMIC, should also be added to monitor charging status during energy generation. Using a PMIC would also allow adding a micro-USB port for charging by external electrical power when it

is available. The usage of a DC-DC converter instead of a series resistor would resolve the issue of not having reached 100% brightness after 5 minutes of cranking in a worst-case scenario, as mentioned in section 4.1, since DC-DC converter can set the output voltage at 3 V regardless of the input voltage. A schematic of a modified circuit diagram, including a PMIC (BQ24010DRC from Texas Instruments) and a DC-DC converter (XCL206A303CR-G from Torex Semiconductor Ltd.), is shown in Figure 13.

The replacement module's mechanical parts also need modification to ensure durability. The hinge and crank arm could be machined out of steel instead of aluminum for greater strength. The gearbox also needs to be redesigned to reliably withstand rough use and erratic cranking, since it is currently too sensitive to angular accelerations during cranking. This could be accomplished by integrating the gearbox with the body of the module using a planetary gear system. Another approach is to limit the angular acceleration of the crank by adding a flexure around the generator shaft, which would attenuate high accelerations.

Finally, the device needs to be re-engineered for manufacturing and assembly on a larger scale.

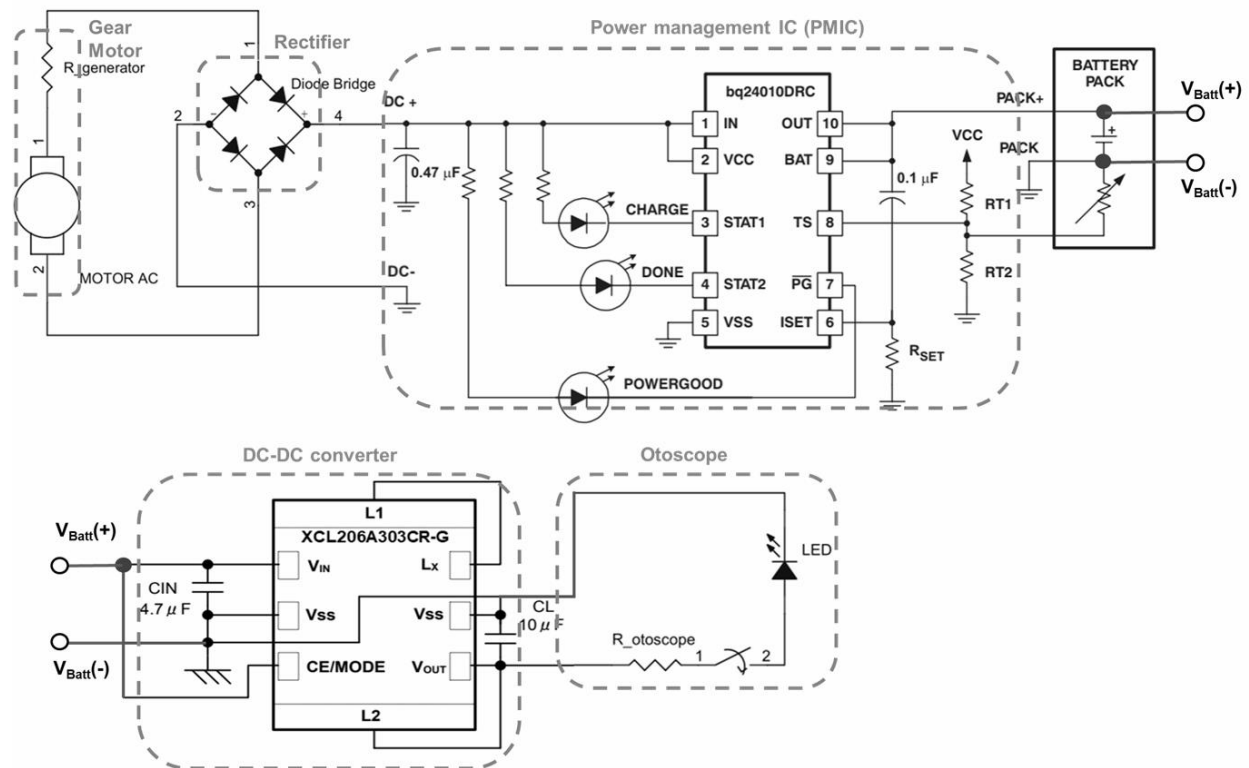


Figure 13. A circuit diagram for future prototypes, including a DC-DC converter and PMIC for better efficiency and brightness/charging regulation.

5. CONCLUSIONS

A functional prototype of a crank-powered rechargeable battery replacement for LED otoscopes was developed. Initial testing of the prototype proves a crank-powered otoscope is feasible and practical. The battery replacement module is capable of powering a full day's worth of patient examinations with only five minutes of comfortable cranking. Although the otoscope brightness achieved with the replacement module ranged between 50-67% of that achieved with two C-cell batteries, some modifications to the module's electronics could increase the brightness achieved to 100%. Future mechanical modifications, such as redesigning the gearbox for the motor, can make the battery replacement module more durable, suitable for long-term use by clinicians during travel or in environments outside of a traditional office.

Though this device was designed and tested for use in otoscopes, the 2 C-cell-shaped form factor enables this device to be adapted for powering similar medical devices, such as ophthalmoscopes. Future development could lead to a robust power source for a variety of medical devices in low-resource areas.

6. ACKNOWLEDGMENTS

The authors would like to thank our communications instructor, Dave Custer, and course administrative assistant, Irina Gaziyeve, for their help and support throughout the project. The authors would also like to thank students in the 2.75 Medical Device Design class for their feedback throughout the course.

7. REFERENCES

- [1] "Deafness and hearing loss." World Health Organization. February 2017. Accessed December 17, 2017. <http://www.who.int/mediacentre/factsheets/fs300/en/>.
- [2] Lindeman, Todd. "Without electricity, 1.3 billion are living in the dark." The Washington Post, November 6, 2015. Accessed December 17, 2017. www.washingtonpost.com/graphics/world/world-without-power/.
- [3] "Energy." ONE. Accessed December 17, 2017. <https://www.one.org/international/issues/energy/>.

- [4] Provost, Claire. "Energy Poverty Deprives 1 Billion of adequate healthcare, says report ." The Guardian, March 7, 2013. Accessed December 17, 2017. <https://www.theguardian.com/global-development/2013/mar/07/energy-poverty-deprives-billion-adequate-healthcare>.
- [5] Brock, Nelson, Caitlin Mackey, Nevan Hanamura, and Michael Fuenfer. "A Hand-Crank Otoscope for Remote Medical Clinics." Final Paper, Massachusetts Institute of Technology. n.d.
- [6] "Student's dynamo-powered otoscope delights doctors." Vanderbilt University. February 24, 2009. Accessed December 17, 2017. <https://engineering.vanderbilt.edu/news/2009/student%E2%80%99s-dynamo-powered-otoscope-delights-doctors/>.
- [7] Gupta, Anshuman. "Need for a Hand-Crank Powered Otoscope in Rural Areas." Final Paper, Johns Hopkins University, 2017.
- [8] Fuenfer, Michael. "A Self-Powered Otoscope." Proposal, Massachusetts Institute of Technology, 2017.
- [9] "Lithium-Ion Battery Maintenance Guidelines, Tektronix.
<http://www.newark.com/pdfs/techarticles/tektronix/LIBMG.pdf>
- [10] "78:1 Metal Gearmotor 20Dx43L mm 6V CB," Pololu. <https://www.pololu.com/product/3703>
- [11] "Light Sensor," Vernier.: <https://www.vernier.com/products/sensors/light-sensors/lis-bta/>
- [12] Morse, Brian. "Some Important Properties of Human Vision." Presentation, Brigham Young University, n.d.